

From AI Pods to AIOps

cisco Live !

Mastering Enterprise AI Workloads in the Modern Data Centre

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Webex App

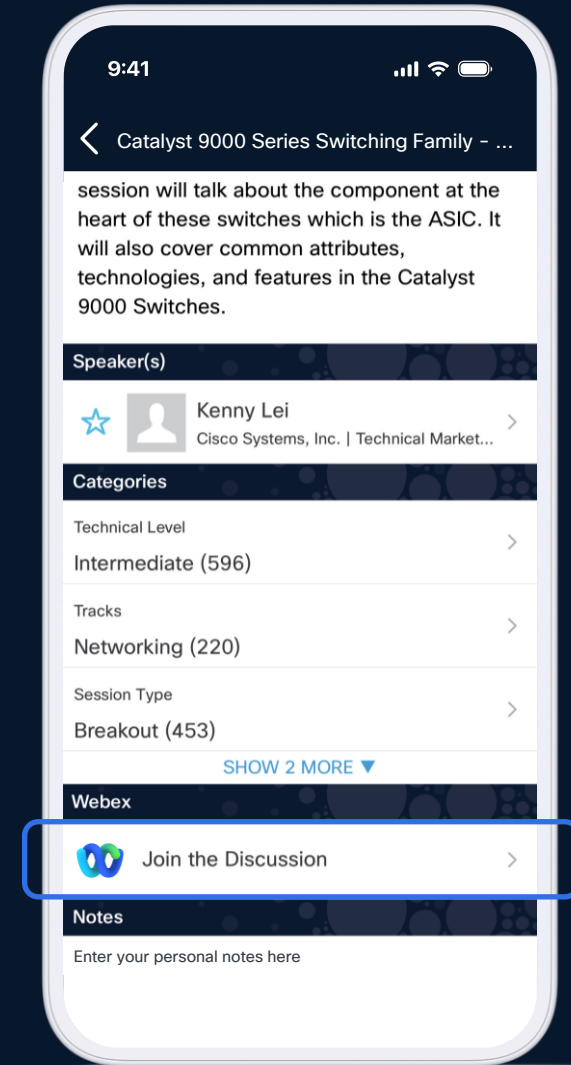
Questions?

Use Webex App to chat with the speaker after the session

How

- 1 Find this session in the Cisco Events App
- 2 Click “Join the Discussion”
- 3 Install the Webex App or go directly to the Webex space
- 4 Enter messages/questions in the Webex space

Webex spaces will be moderated by the speaker until February 27, 2026.



Meet Your Presenters

Antonio Gallego



AI Solutions Engineer,
Telco & Neocloud, EMEA



Madrid, Spain



Patent holder in <1 year at
Cisco, 40 years of gaming



Jara Osterfeld

AI Solutions Engineer,
Enterprise, Northern Europe



Rotterdam, The Netherlands



Part of the Cisco Live NOC,
training for my first triathlon

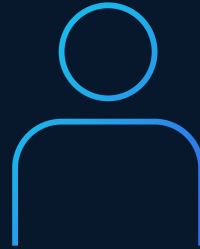


Meet the AICME Team

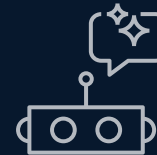
Antonio Gallego



**AI Infra Manager,
Platform Admin**



keeps the lights on,
ensures AI workloads
run efficiently,
provides developers
with what they need



Jara Osterfeld



**AI Developer,
Platform Consumer**

deploys and maintains
AI applications and
agents, used to worry
about infrastructure
and performance

Agenda

09:00	Standup Call
09:30	Incident Response
11:00	Deployment and Security
14:00	Optimization and Scaling
16:00	Governance and Compliance
17:30	Packing Up

09:00

Stand Up Call

6 months ago ...



Troubleshooting
infrastructure issues



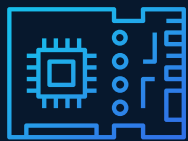
No visibility into
jobs scheduling



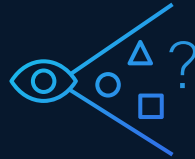
Model failure
in production



Cognitive
Load



Inconsistent
GPU utilization



No visibility into
model performance
until production fails



Security concerns
with model
inputs/outputs



Manual
processes



The challenges we had



Volume of Data



150k
IT users



100+
services



300k
endpoints



21k
apps



60k+
VMs



150k
containers



24k
APs



8k LAN
switches



2k+
routers



11k
servers



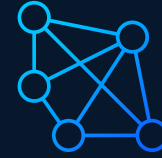
3k
DBs



300k
nodes



Operational Complexity



Dependencies



1.5M
events/day



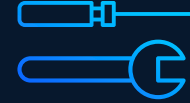
23 TB
logs/day



50+
integrations



500+
KPIs



Number of Tools



infrastructure
management



security

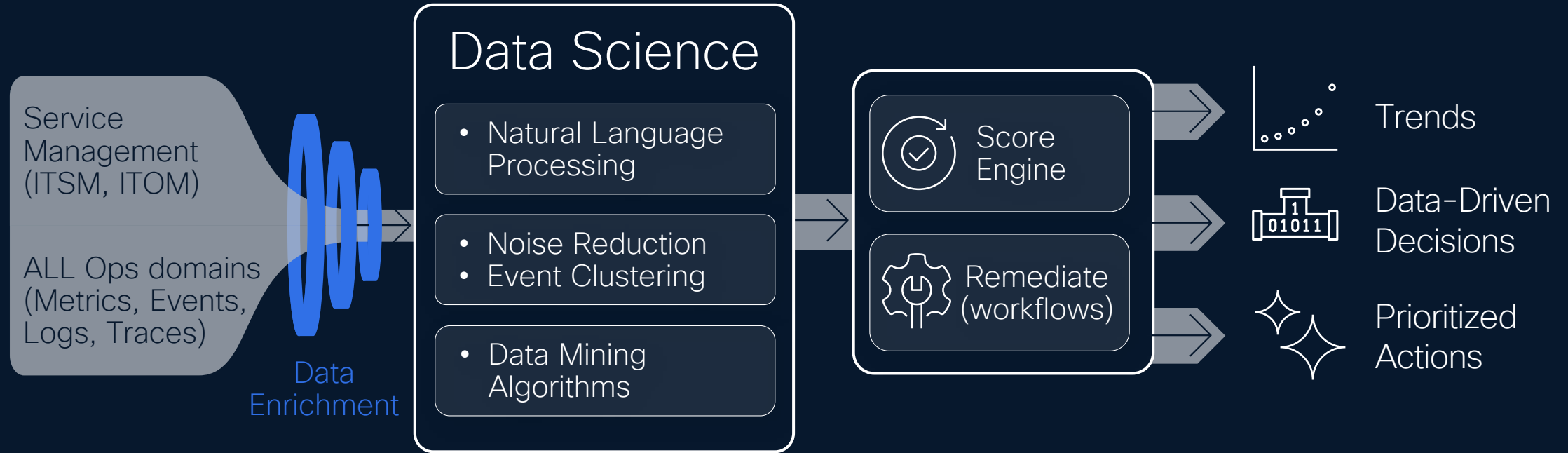


observability



automation

The goal in mind: Operational Excellence with AIOps



Noise Reduction
De-duplication

Event Clustering
Decision-making

Predictive Analytics
Model training

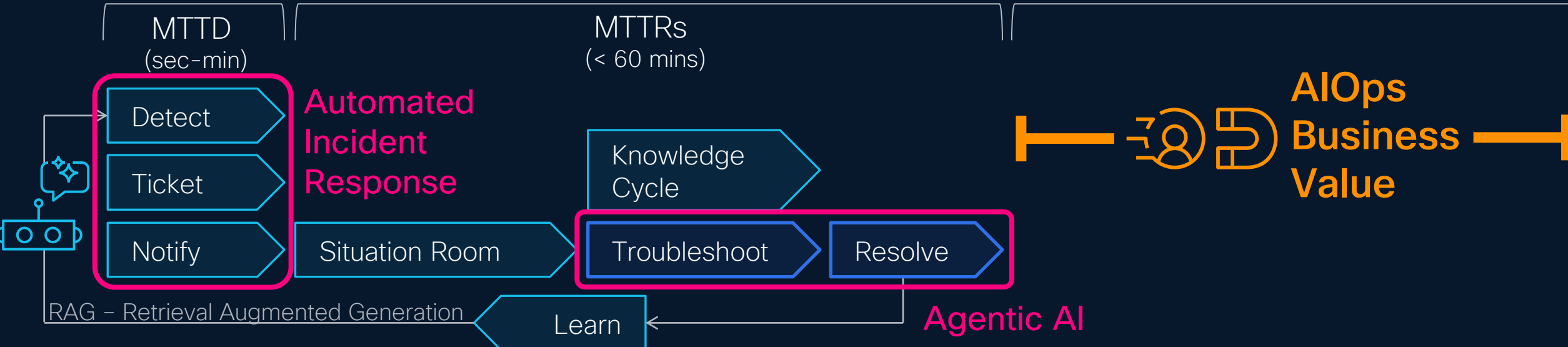
Change Assessment
Change risk scoring

The promise to management

Before AIOps



After AIOps



The critical questions management asked

Before AIOps

MTTD (16 mins)

MTTRs (104 mins)

MTTD (30 mins)

Angonised → Correlate → Ticket → Nitge → Bridge → Troubleshoot → Restore → Resolve

After AIOps

MTTD (sec-mins)

MTTRs (< 60 mins)

Detect → Ticket → Notify

Automated Incident Response

Situation Rooms

Knowledge Cycle

Troubleshoot → Resolve

AIOps Business Value

Can I trust AI enough to make decisions for our business?

Are we amplifying our best intellectual property – or our biggest data liabilities?

Looks great in a demo — but will it survive a Monday morning spike?

The choices we needed to make

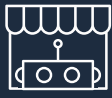
Agent Ecosystem



Chatbot Applications



AI Agents



Agent Hub



Agent Platform

AI Platform Services

Foundational Models

Open or closed



Custom Models

Pre-training, Fine-tuning



Model Serving and Deployment

Registries



Model and Agent Tracing

AI guardrails



Data Infrastructure

Vector, Ingestion, Embedding, Retrieval



Infrastructure Platform

AI-Ready Infrastructure



Resource Management and Workload Orchestration



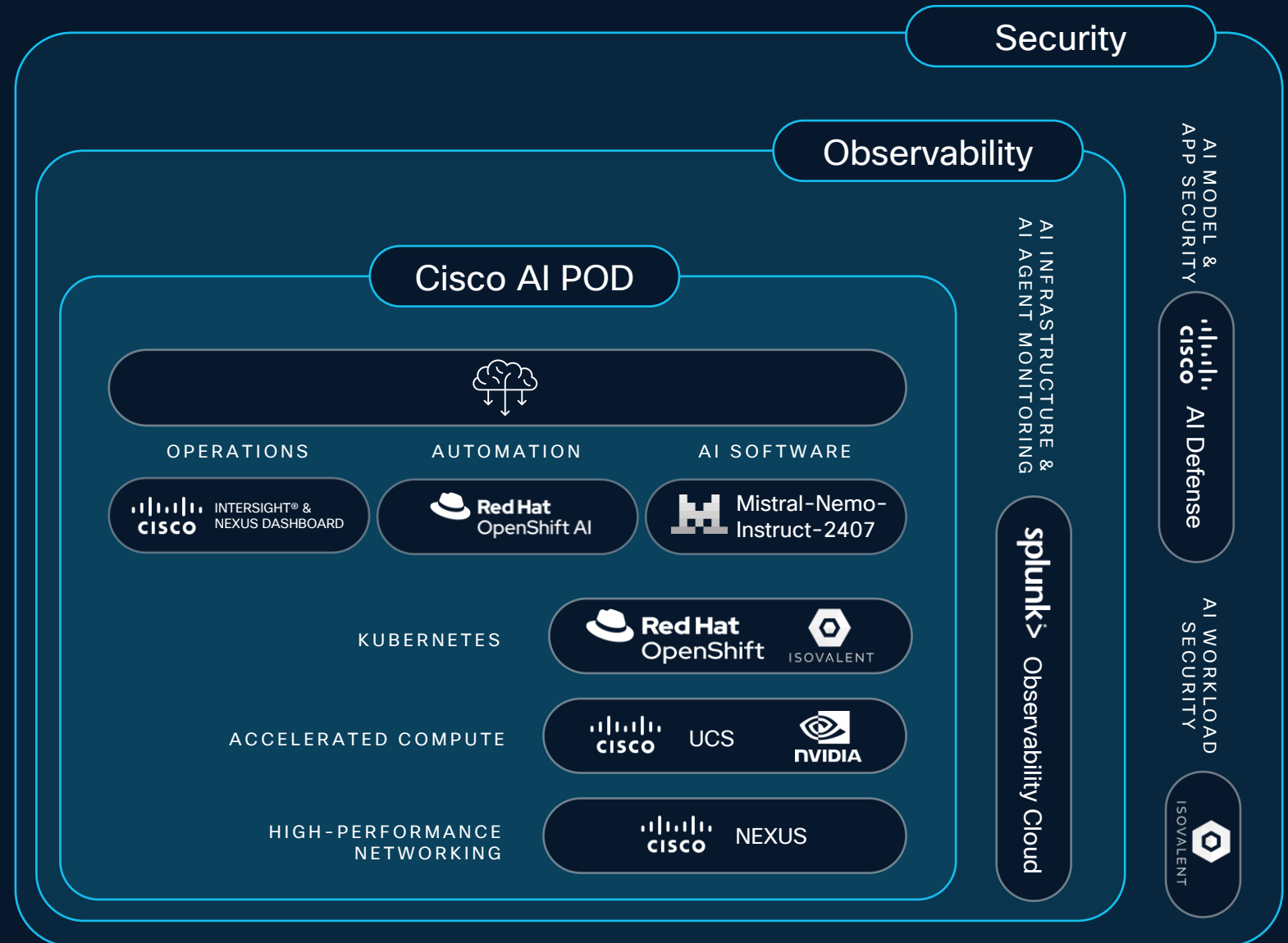
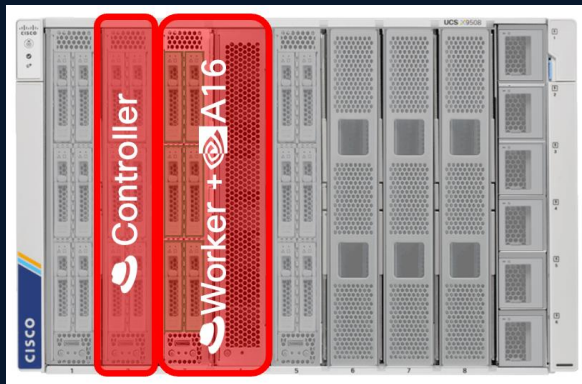
Multi-Tenant GPUaaS



The solution we built

Cisco Secure AI Factory with NVIDIA

- Single-Node bare metal OpenShift + 1x worker node
- UCSX M6 nodes, 1x A16 GPU
- Connected to Intersight and Nexus Dashboard



09:30

Incident Response

The Challenge

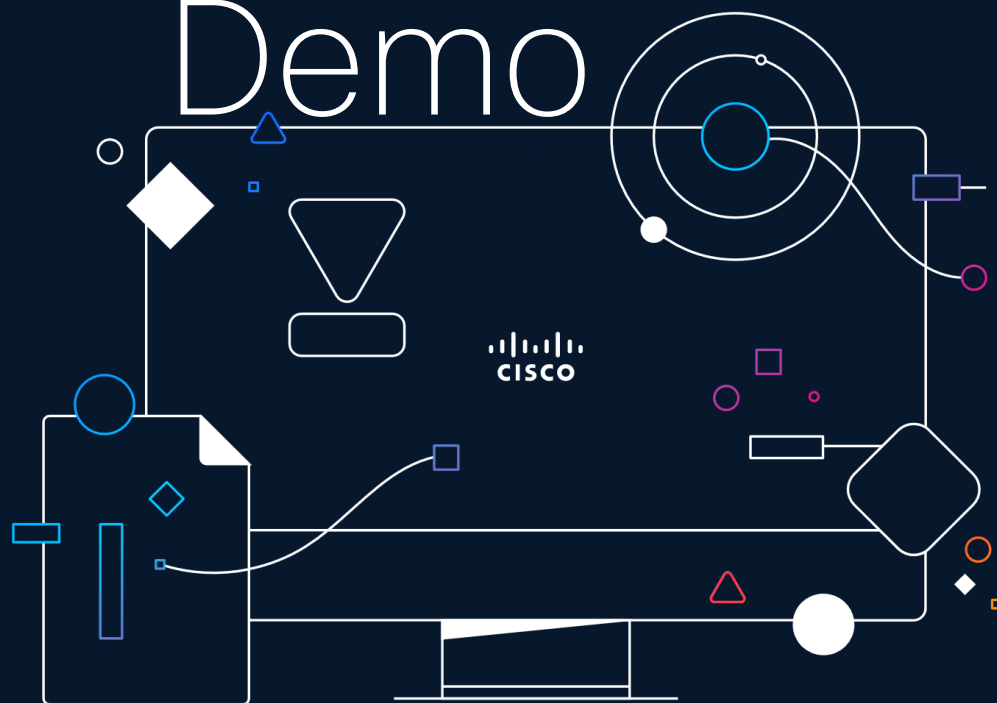


Production AI services experience **performance degradation** and it is impacting **customer experience** and **SLA compliance**.

Traditional approaches require **manual investigation** across **disconnected monitoring tools**, leading to **extended downtime** and **resource waste**.



Demo



The Result

Integrated observability through Splunk monitoring of Cisco AI Pod infrastructure enables rapid root cause identification and automated remediation. The platform detected thermal throttling on a GPU node affecting model inference latency (120ms → 380ms) and correlated hardware metrics with application performance in real-time.



Business Outcomes

- **MTTR Reduction:** Incident resolution time reduced from hours to 6 minutes
- **Zero Application Changes:** Workload migration handled transparently by OpenShift AI
- **Proactive Operations:** Issue detected and resolved before customer impact
- **Cost Avoidance:** Prevented potential SLA breach and associated penalties



ROI Indicators

- 95% reduction in incident response time
- 100% SLA maintenance during infrastructure events
- Eliminated emergency weekend maintenance windows



Key Technology Integrations

Splunk Observability Cloud + Cisco AI Pod telemetry + OpenShift AI orchestration

11:00

Deployment and Security

The Challenge

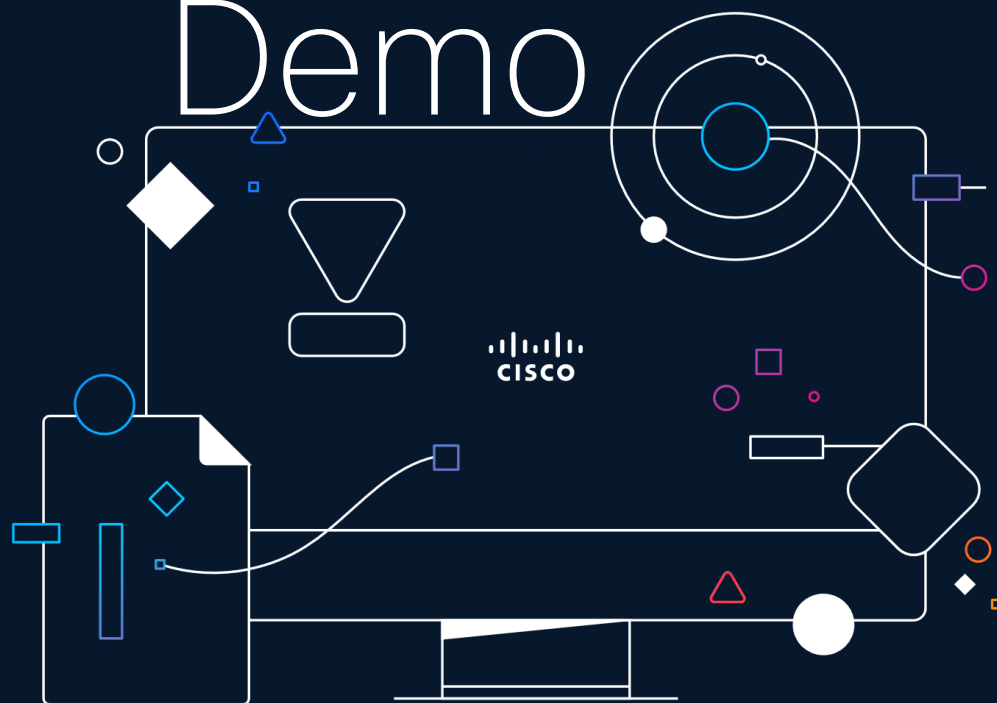


There is a dual challenge of accelerating **AI deployment velocity** while ensuring **robust security and compliance**.

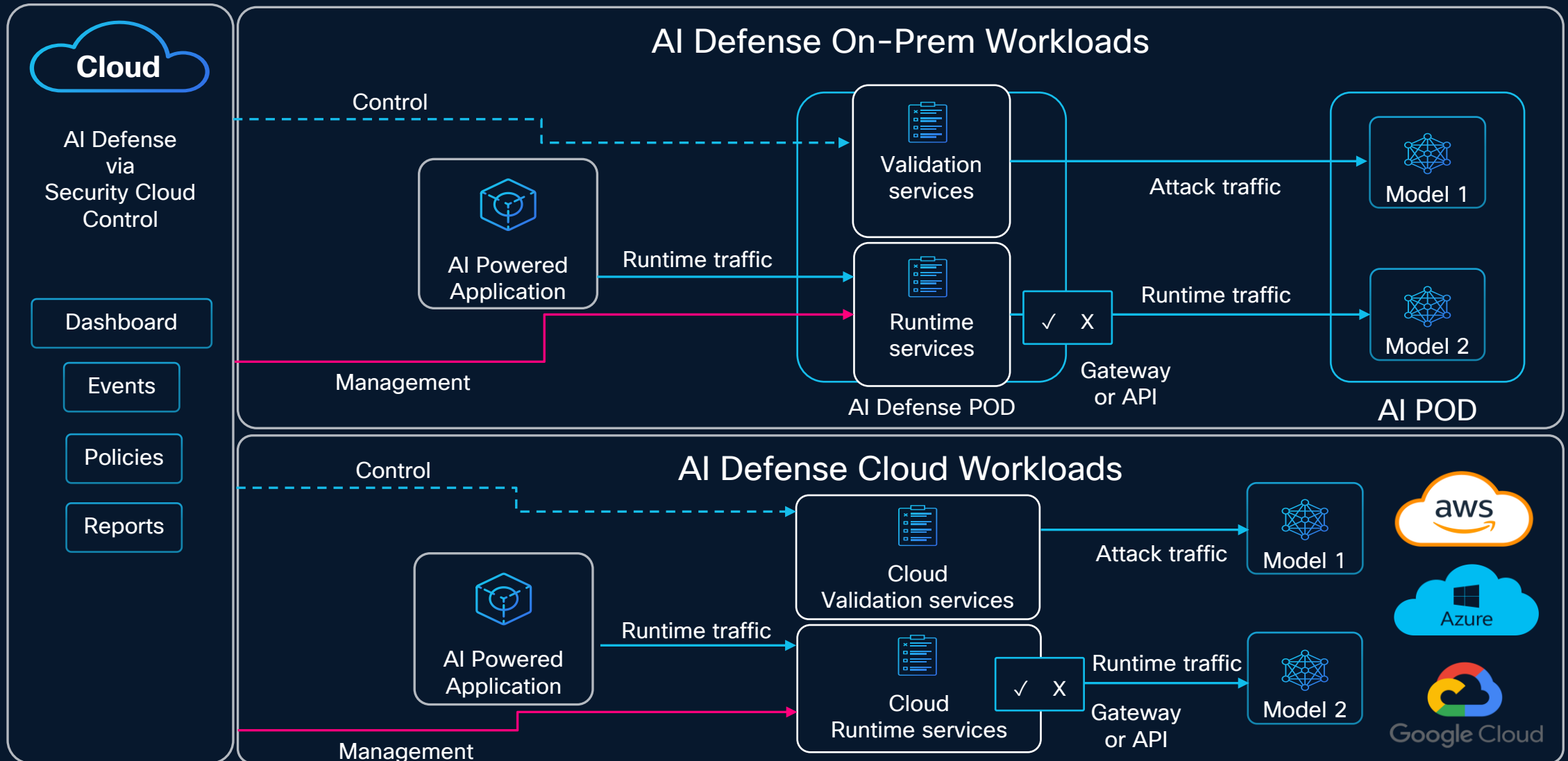
Deploying AI models into production traditionally involves **lengthy change management processes**, **security review delays**, and exposure to **AI-specific threats** like prompt injection, data leakage, and model manipulation.



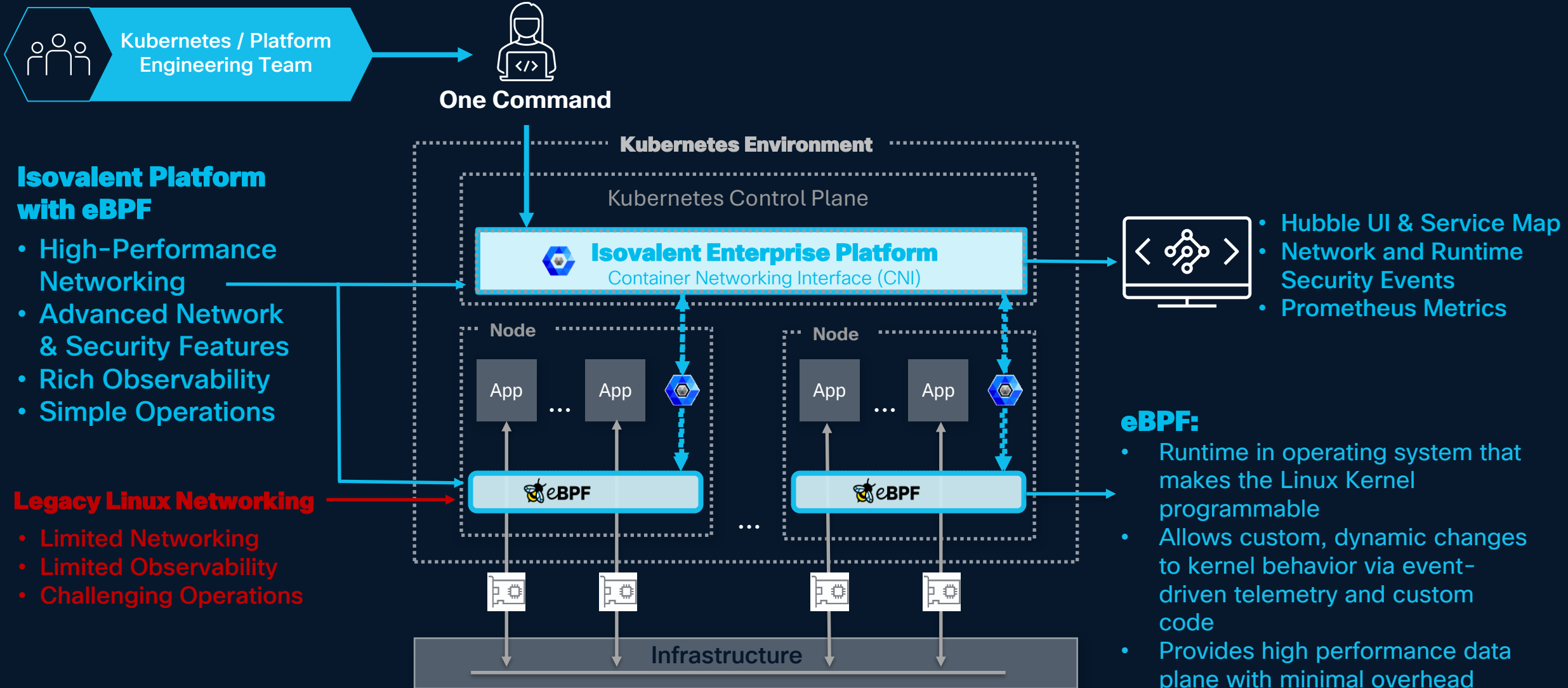
Demo



Cisco AI Defense



Isovalent: AI Networking and Security in Kubernetes



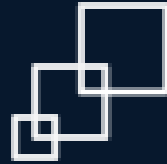
Isovalent: AI Networking and Security in Kubernetes



Networking
/ Routing



Firewall /
Microseg



Load
Balancing



Network
Visibility



VPN /
Encryption



Runtime
Security

Why does this matter for AI Workloads?

- Large Kubernetes environments are often comprised of **1,000s of nodes with high churn** to maximize GPU utilization.
- Transferring **very large data sets, models, training data, etc** requires high-throughput.
- Some use cases (e.g., real-time inference) are **very latency sensitive**.

How do eBPF / Isovalent help?

- **Designed for high-scale & churn** with incremental state updates.
- Highly-optimized in-kernel data path for **high-throughput + low latency**.
- Seamlessly **interconnect multiple clusters** to achieve massive scale.

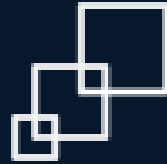
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Security

Why does this matter for AI Workloads?

- Training / inference data is often **highly sensitive**.
- Models represent **critical intellectual property**.
- Environments are commonly **multi-tenant** to maximize utilization of \$\$ GPUs.
- Risk of **data exfiltration** from compromise or insider attacks is significant.

How do eBPF / Isovalent help?

- Provides **identity-aware per-tenant isolation**, even with high scale & churn.
- DNS-aware policy allows **fine-grained per-tenant egress controls**.
- L7+TLS policy rules allow for **deep inspection + control of HTTP data flows**

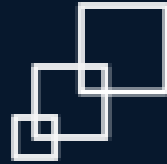
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Security

Why does this matter for AI Workloads?

- **Auto-scaled services** within the cluster requires efficient + distributed L3/L4 LB.
- **External access to UI/APIs** for research, inference, etc. requires north/south L7 LB with auth + auto-scaling.
- **High-bandwidth inference** (e.g., video) needs efficient N/S L3/L4 LB

How do eBPF / Isovalent help?

- **In-kernel east-west service load-balancing** is more scalable than iptables and drastically more efficient than a traditional proxy.
- Seamlessly integrates **Envoy for L7 north-south LB**.
- XDP enables **commodity HW to provide NIC accelerated L3/L4 LB**

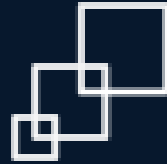
Isovalent: AI Networking and Security in Kubernetes



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Runtime
Security

Why does this matter for AI Workloads?

- A **broken AI** app deployed to a large set of GPU nodes without creating useful output is **EXTREMELY expensive**.
- The risk of a broken app increases as network firewalling is introduced.

How do eBPF / Isovalent help?

- **Isovalent Networking for Kubernetes** provides **identity-aware network observability**.
- **Lightweight eBPF** network observability can **see connectivity faults at all layers** (L2 through L3/L4 and L7 protocols like HTTP).
- Helps speed **root cause analysis and MTTR**.

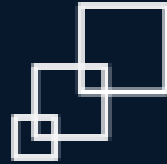
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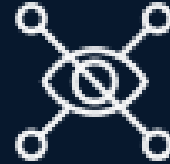
Networking
/ Routing



Firewall /
Microseg



Load
Balancing



Network
Visibility



VPN /
Encryption



Runtime
Security

Why does this matter for AI Workloads?

- Both **training and inference data** often include **sensitive data** subject to **compliance requirements** that require **encryption** for data in flight.
- The **wide variety of AI frameworks** makes securing this data at the application layer difficult to do this reliably.

How do eBPF / isovalent help?

- **Transparent encryption.** It encrypts traffic "in-transit" between pods and nodes within a cluster automatically, without needing any changes to the applications themselves.
- **Kernel-level encryption methods** like IPsec and WireGuard provide encryption with **high-performance**.

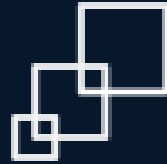
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Runtime
Security

Why does this matter for AI Workloads?

- Workloads have access to incredibly **sensitive data & models**.
- Workloads, particularly those for research, run **unvetted 3rd-party code / libs**.
- Risk of **data exfiltration** by code compromise or insider attack is high.

How do eBPF / Isovalent help?

- **Runtime security telemetry** for processes, files and network connections.
- Data can be exported to a SIEM for **attack detection**.
- Tetragon can **enforce policies** (e.g., kill any process accessing the internet after reading the model file).

The Result

Self-service model deployment through OpenShift AI combined with AI-native security enforcement via Cisco AI Defense and Isovalent. Developer deploys a customer-facing LLM chatbot in under 5 minutes with automated security guardrails that detect and block prompt injection attempts, PII exposure, and toxic content in real-time.



Business Outcomes

- **Deployment Velocity:** Model-to-production time reduced from 2-3 weeks to 5 minutes
- **Security Automation:** 100% of AI-specific threats blocked without manual intervention
- **Compliance Assurance:** Automated PII detection and redaction ensures regulatory compliance
- **Developer Productivity:** 60% time savings by eliminating security review bottlenecks



ROI Indicators

- 99.8% reduction in deployment cycle time
- Zero AI security incidents since implementation
- 40% increase in AI model deployment frequency
- Automated compliance for GDPR, HIPAA, PCI-DSS requirements



Key Technology Integrations

OpenShift AI model serving + Cisco AI Defense + Isovalent + vLLM inference runtime

14:00

Optimization and Scaling

The Challenge

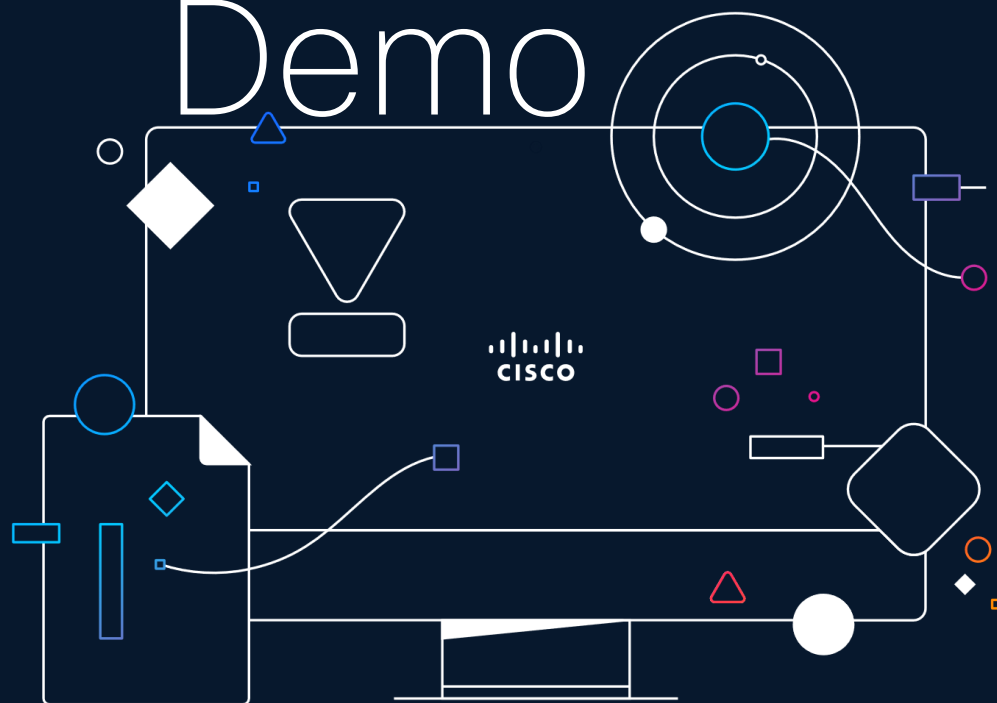


There is a struggle with **AI infrastructure cost management** and **capacity planning**. Teams either over-provision resources (wasting budget) or under-provision (causing performance issues). SLA

Lack of visibility into resource consumption, utilization patterns, and cost allocation leads to **unpredictable cloud bills** and **inefficient resource usage** across multiple teams.



Demo



The Result

Comprehensive resource analytics combining Cisco AI Pod infrastructure metrics, OpenShift AI orchestration data, and application performance indicators through Splunk. Platform provides real-time cost visibility (\$4.80/hour GPU consumption), predictive capacity planning (15% week-over-week growth trend), and automated scaling policies based on performance and cost.



Business Outcomes

- **Cost Optimization:** 35% reduction in GPU infrastructure costs through right-sizing
- **Capacity Predictability:** Data-driven forecasting enables proactive infrastructure planning
- **Multi-Tenancy Efficiency:** Resource quotas eliminate team conflicts and ensure fair allocation
- **Automated Scaling:** Platform scales from 1 to 5 replicas based on demand without manual intervention



ROI Indicators

- 35% GPU cost reduction (\$42K/month savings at enterprise scale)
- 80% reduction in resource contention incidents
- 5-minute self-service quota adjustment vs. 5-day traditional procurement
- 99.7% resource utilization within defined quotas



Key Technology Integrations

Splunk AI workload analytics + OpenShift AI horizontal pod autoscaling + Cisco AI Pod capacity management

16:00

Governance and Compliance

The Challenge



Enterprise AI initiatives face increasing **regulatory scrutiny** and **audit requirements** (SOC 2, GDPR, HIPAA, ISO 27001).

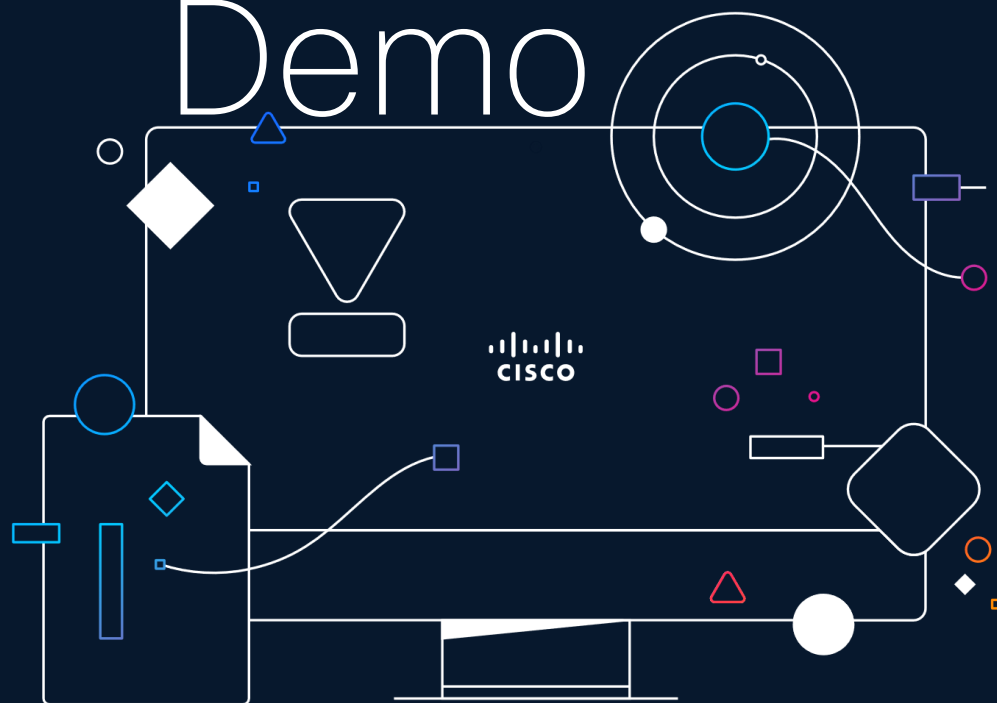
Organizations need to prove:

- What models are deployed?
 - Who deployed them?
 - How is data protected?
- Are security policies are enforced?

Traditional compliance processes require **manual evidence collection across disconnected systems**, consuming significant operational resources and delaying audit responses.



Demo



The Result

Unified audit trail aggregating governance data from OpenShift AI (deployment history, access controls), Cisco AI Defense (security incidents, policy enforcement), and Splunk (performance SLA compliance, change management). A **comprehensive compliance report** that previously required 2 days of manual work is now generated automatically in 2 minutes.



Business Outcomes

- **Audit Efficiency:** Compliance report generation time reduced from 2 days to 2 minutes (99% time savings)
- **Continuous Compliance:** Real-time policy enforcement eliminates compliance gaps
- **Reduced Risk:** Complete audit trail provides forensic capabilities and accountability
- **Accelerated Certifications:** Automated evidence collection speeds certification processes



ROI Indicators

- 99% reduction in compliance report preparation time
- Zero compliance violations since implementation
- 60% reduction in audit preparation costs (\$150K+ annually)
- 4-6 weeks faster security certification timeline



Key Technology Integrations

OpenShift AI audit logging + Cisco AI Defense compliance policies + Splunk unified audit dashboard

17:30

Packing Up

AI requires more than “just” GPUs



Trust

Control data, privacy and speed

Choice

Open, curated, and validated technology stack

Consistency

One way to build, run, and access applications and agents

Where we are today



Accelerated
Incident
Resolution



Seamless
Operations
and Reliability



Rapid and
Secure Model
Deployment



Optimized
Resource Utilization
and Cost Savings



Automated
Compliance and
Audit Efficiency



Mature AI operations aren't about having the best models – they're about having the best platform. Organizations that master AIOps will dominate their markets; those that don't will struggle to move beyond pilot projects.

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